



CONTENTS IN DETAIL

ACKNOWLEDGMENTS

INTRODUCTION

Who This Book Is For

What's in This Book

Python Version, Platform, and IDE

The Code

Coding Style

Where to Get Help

Onward!

1

SILLY NAME GENERATOR

Project #1: Generating Pseudonyms

Planning and Designing a Project

The Strategy

The Pseudocode

The Code

Using the Python Community's Style Guide

Checking Your Code with Pylint

Describing Your Code with Docstrings

Checking Your Code Style

Summary

Further Reading

Pseudocode

Style Guides

Third-Party Modules

Practice Projects

Pig Latin

Poor Man's Bar Chart

Challenge Projects

Poor Foreign Man's Bar Chart

The Middle

Something Completely Different

2

FINDING PALINGRAM SPELLS

Finding and Opening a Dictionary

Handling Exceptions When Opening Files

Loading the Dictionary File

Project #2: Finding Palindromes

The Strategy and Pseudocode

The Palindrome Code

Project #3: Finding Palingrams

The Strategy and Pseudocode

The Palingrams Code

Palingram Profiling

Palingram Optimization

dnE ehT

Further Reading

Practice Project: Dictionary Cleanup

Challenge Project: Recursive Approach

3

SOLVING ANAGRAMS

Project #4: Finding Single-Word Anagrams

The Strategy and Pseudocode

Anagram-Finder Code

Project #5: Finding Phrase Anagrams

The Strategy and Pseudocode

The Anagram Phrase Code

Project #6: Finding Voldemort: The Gallic Gambit

Project #7: Finding Voldemort: The British Brute-Force

Strategy

The British Brute-Force Code

Summary

Further Reading

Practice Project: Finding Digrams

Challenge Project: Automatic Anagram Generator

4

DECODING AMERICAN CIVIL WAR CIPHERS

Project #8: The Route Cipher

The Strategy

The Pseudocode

The Route Cipher Decryption Code

Hacking the Route Cipher

Adding a User Interface

Project #9: The Rail Fence Cipher

The Strategy

The Rail Fence Cipher Encryption Code

The Rail Fence Cipher Decryption Code

Summary

Further Reading

Practice Projects

Hacking Lincoln

Identifying Cipher Types

Storing a Key as a Dictionary

Automating Possible Keys

Route Transposition Cipher: Brute-Force Attack

Challenge Projects

Route Cipher Encoder

Three-Rail Fence Cipher

5

ENCODING ENGLISH CIVIL WAR CIPHERS

Project #10: The Trevanion Cipher

Strategy and Pseudocode

The Trevanion Cipher Code

Project #11: Writing a Null Cipher

The List Cipher Code

The List Cipher Output

Summary

Further Reading

Practice Projects

Saving Mary

The Colchester Catch

6

WRITING IN INVISIBLE INK

Project #12: Hiding a Vigenère Cipher

The Platform

The Strategy

Creating Invisible Ink

Manipulating Word Documents with python-docx

Downloading the Assets

The Pseudocode

The Code

Importing python-docx, Creating Lists, and Adding a Letterhead

Formatting and Interleaving the Messages

Adding the Vigenère Cipher

Detecting the Hidden Message

Summary

Further Reading

Practice Project: Checking the Number of Blank Lines

Challenge Project: Using Monospace Font

7

BREEDING GIANT RATS WITH GENETIC ALGORITHMS

Finding the Best of All Possible Solutions

Project #13: Breeding an Army of Super-Rats

Strategy

The Super-Rats Code

Summary

Project #14: Cracking a High-Tech Safe

Strategy

The Safecracker Code

Summary

Further Reading

Challenge Projects

Building a Rat Harem

Creating a More Efficient Safecracker

8

COUNTING SYLLABLES FOR HAIKU POETRY

Japanese Haiku

Project #15: Counting Syllables

The Strategy

Using a Corpus

Installing NLTK

Downloading CMUdict

Counting Sounds Instead of Syllables

Handling Words with Multiple Pronunciations

Managing Missing Words

The Training Corpus

The Missing Words Code

The Count Syllables Code

Prepping, Loading, and Counting

Defining the main() Function

A Program to Check Your Program

Summary

Further Reading

Practice Project: Syllable Counter vs. Dictionary File

Project #16: Markov Chain Analysis

The Strategy

Choosing and Discarding Words

Continuing from One Line to Another

The Pseudocode

The Training Corpus

Debugging

Building the Scaffolding

Using the logging Module

The Code

Setting Up

Building Markov Models

Choosing a Random Word

Applying the Markov Models

Generating the Haiku Lines

Writing the User Interface

The Results

Good Haiku

Seed Haiku

Summary

Further Reading

Challenge Projects

New Word Generator

Turing Test

Unbelievable! This Is Unbelievable! Unbelievable!

To Haiku, or Not to Haiku

Markov Music

10

ARE WE ALONE? EXPLORING THE FERMI PARADOX

Project #17: Modeling the Milky Way

The Strategy

Estimating the Number of Civilizations

Selecting Radio Bubble Dimensions

Generating a Formula for the Probability of Detection

The Probability-of-Detection Code

Calculating Probability of Detection for a Range of Civilizations

Generating a Predictive Formula and Checking the Results

Building the Graphical Model

Scaling the Graphical Model

The Galaxy Simulator Code

Results

Summary

Further Reading

Practice Projects

A Galaxy Far, Far Away

Building a Galactic Empire

A Roundabout Way to Predict Detectability

Challenge Projects

Creating a Barred-Spiral Galaxy

Adding Habitable Zones to Your Galaxy

11

THE MONTY HALL PROBLEM

Monte Carlo Simulation

Project #18: Verify vos Savant

The Strategy

The vos Savant Verification Code

Project #19: The Monty Hall Game

A Brief Introduction to Object-Oriented Programming

The Strategy and Pseudocode

Game Assets

The Monty Hall Game Code

Summary

Further Reading

Practice Project: The Birthday Paradox

12

SECURING YOUR NEST EGG

Project #20: Simulating Retirement Lifetimes

The Strategy

Historical Returns Matter

The Greatest Uncertainty

A Qualitative Way to Present Results

The Pseudocode

Finding Historical Data

The Code

Importing Modules and Defining Functions to Load Data and Get User Input

Getting the User Input

Checking for Other Erroneous Input

Defining the Monte Carlo Engine

Simulating Each Year in a Case

Calculating the Probability of Ruin

Defining and Calling the main() Function

Using the Simulator

Summary

Further Reading

Challenge Projects

A Picture Is Worth a Thousand Dollars

Mix and Match

Just My Luck!

All the Marbles

13

SIMULATING AN ALIEN VOLCANO

Project #21: The Plumes of Io

A Slice of pygame

The Strategy

Using a Game Sketch to Plan

Planning the Particle Class

The Code

Importing Modules, Initiating pygame, and Defining Colors

Defining the Particle Class

Ejecting a Particle

Updating the Particle and Handling Boundary Conditions

Defining the main() Function

Completing the main() Function

Running the Simulation

Summary

Further Reading

Practice Project: Going the Distance

Challenge Projects

Shock Canopy

The Fountainhead

With a Bullet

14

MAPPING MARS WITH THE MARS ORBITER

Astrodynamics for Gamers

The Law of Universal Gravity

Kepler's Laws of Planetary Motion

Orbital Mechanics

Project #22: The Mars Orbiter Game

The Strategy

Game Assets

The Code

Importing and Building a Color Table

Defining the Satellite Class Initialization Method

Setting the Satellite's Initial Position, Speed, Fuel, and Sound

Firing Thrusters and Checking for Player Input

Locating the Satellite

Rotating the Satellite and Drawing Its Orbit

Updating the Satellite Object

Defining the Planet Class Initialization Method

Rotating the Planet

Defining the gravity() and update() Methods

Calculating Eccentricity

Defining Functions to Make Labels

Mapping Soil Moisture

Casting a Shadow

Defining the main() Function

Instantiating Objects, Setting Up Orbit Verification, Mapping, and Timekeeping

Starting the Game Loop and Playing Sounds

Applying Gravity, Calculating Eccentricity, and Handling Failure

Rewarding Success and Updating and Drawing Sprites

Displaying Instructions and Telemetry and Casting a Shadow

Summary

Challenge Projects

Game Title Screen

Smart Gauges

Radio Blackout

Scoring

Strategy Guide

Aerobraking

Intruder Alert!

Over the Top

15

IMPROVING YOUR ASTROPHOTOGRAPHY WITH PLANET STACKING

Project #23: Stacking Jupiter

The pillow Module

Working with Files and Folders

Directory Paths

The Shell Utilities Module

The Video

The Strategy

The Code

The Cropping and Scaling Code

The Stacking Code

The Enhancing Code

Summary

Further Reading

Challenge Project: Vanishing Act

16

FINDING FRAUDS WITH BENFORD'S LAW

Project #24: Benford's Law of Leading Digits

Applying Benford's Law

Performing the Chi-Square Test

The Dataset

The Strategy

The Code

Importing Modules and Loading Data

Counting First Digits

Getting the Expected Counts

Determining Goodness of Fit

Defining the Bar Chart Function

Completing the Bar Chart Function

Defining and Running the main() Function

Summary

Further Reading

Practice Project: Beating Benford

Challenge Projects

Benfording the Battlegrounds

While No One Was Looking

APPENDIX

PRACTICE PROJECT SOLUTIONS

Chapter 1: Silly Name Generator

Chapter 2: Finding Palingram Spells

Chapter 3: Solving Anagrams

Chapter 4: Decoding American Civil War Ciphers

Chapter 5: Encoding English Civil War Ciphers

Chapter 6: Writing in Invisible Ink

Chapter 8: Counting Syllables for Haiku Poetry

Chapter 10: Are We Alone? Exploring the Fermi Paradox

Chapter 11: The Monty Hall Problem

Chapter 13: Simulating an Alien Volcano

Chapter 16: Finding Frauds with Benford's Law

INDEX